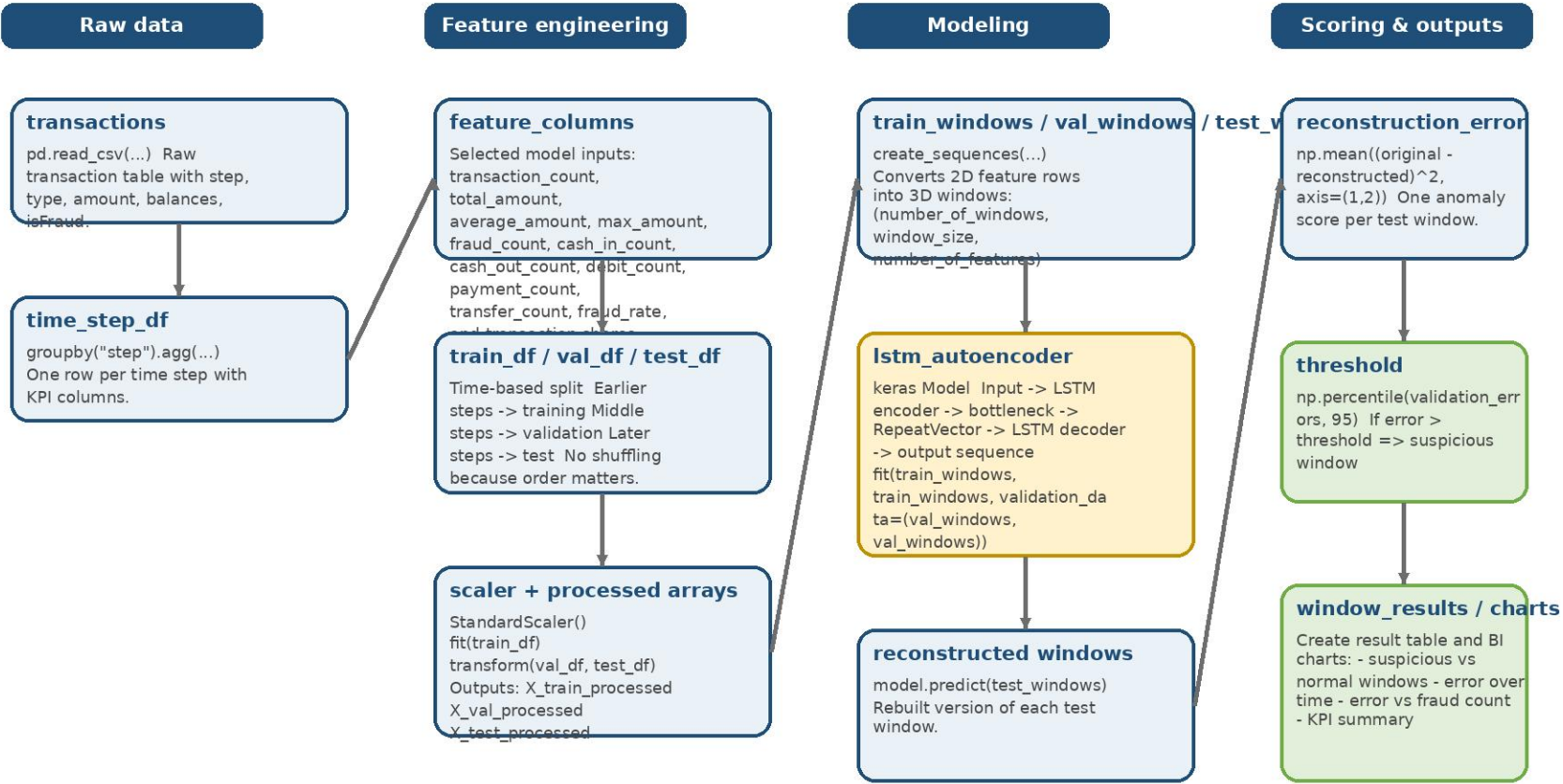


Time-Series Autoencoder - Variable / Method Flow

Flowchart of the main variables and methods used in the notebook



How to read this flow

- Variables are shown as the main named objects created in the notebook.
- Methods are written inside each box to show how the next variable is produced.
- The model is unsupervised during training: it reconstructs input windows, not fraud labels.
- Fraud information is used later for interpretation only, to see whether suspicious windows overlap with risky periods.
- The most important shape change is: rows -> processed feature matrix -> sequence windows -> reconstructed windows -> one error score per window.

Mini glossary

- Window:** A block of consecutive time steps used as one training example.
- Bottleneck:** Compressed hidden representation that stores the important pattern.
- Reconstruction error:** Difference between original sequence and reconstructed sequence.
- Threshold:** Cutoff value used to decide whether a window is suspicious.